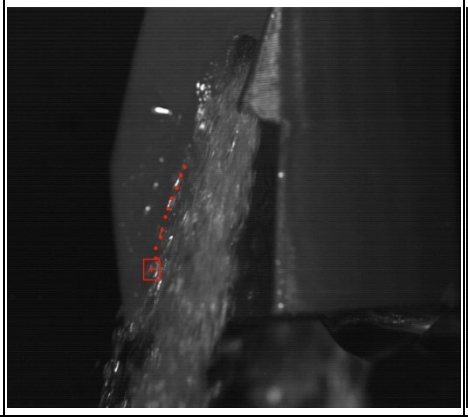
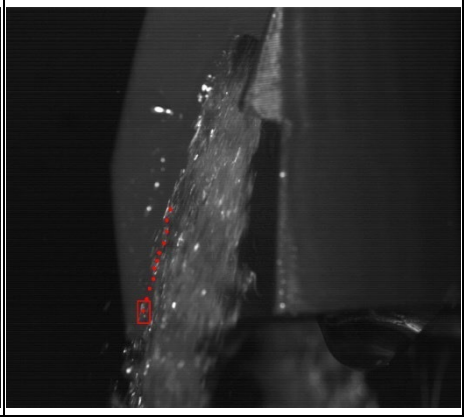
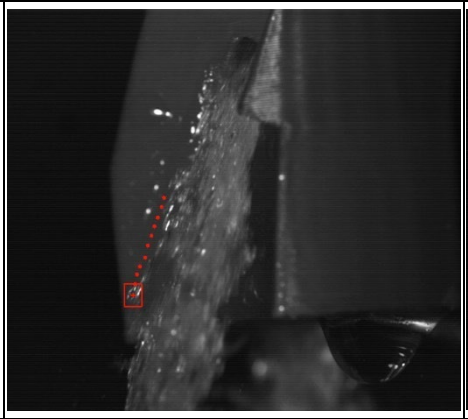
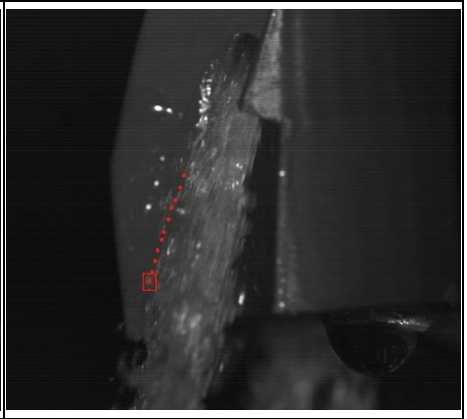
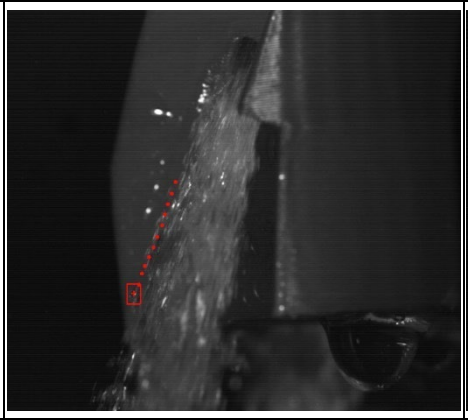
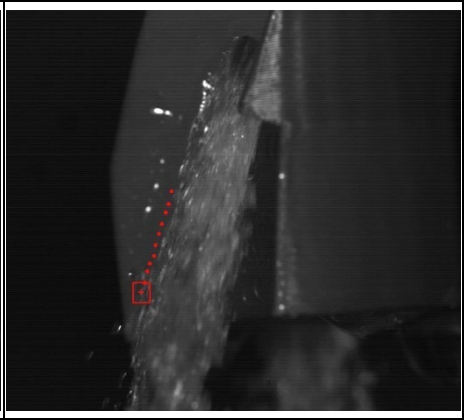
Appendix A: Outlet Flow Velocity Measurement Data at Different Inlet Pressures Before Optimization (1 pixel = 3.85×10-5 m)

Table 8 Pre-optimization outlet velocity measurement data (inlet pressure 0.6MPa)

Type	Start	End	Start	End
Trajectory diagram				
Coordinates	(562,334)	(494,547)	(533,422)	(478,634)
Trajectory diagram				
Coordinates	(520.5,393)	(456,596)	(560,357)	(488,578)
Trajectory diagram				
Coordinates	(544,360.5)	(458,592)	(533.5,382.5)	(472,592)

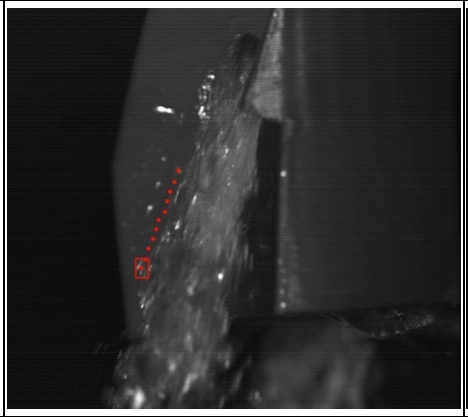
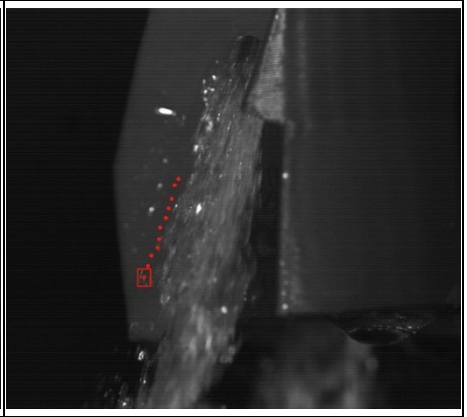
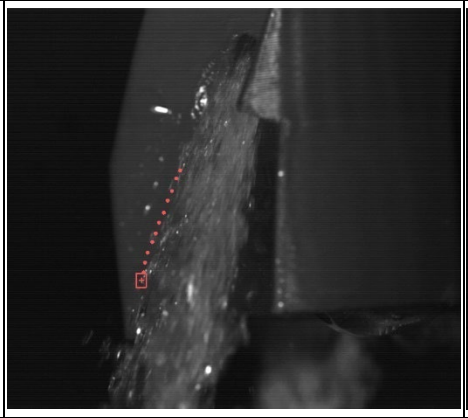
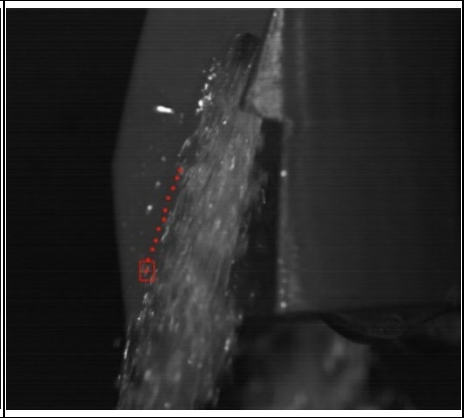
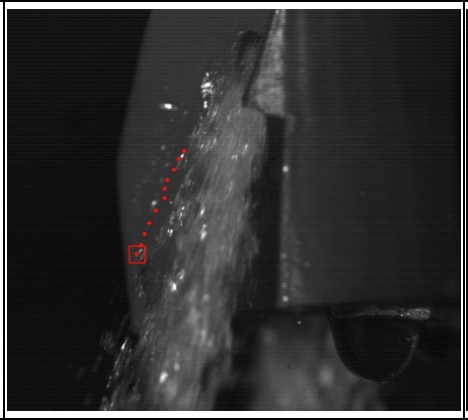
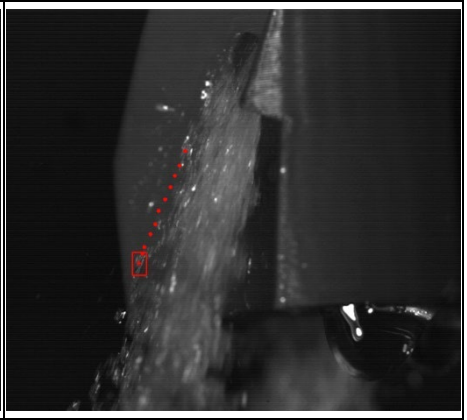
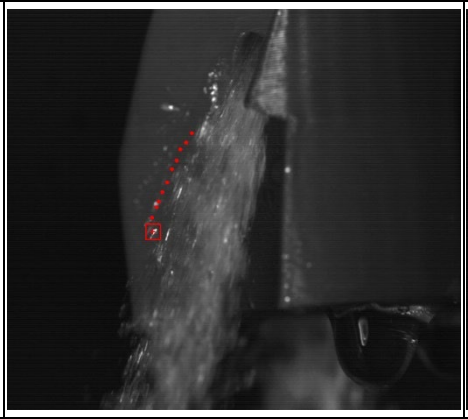
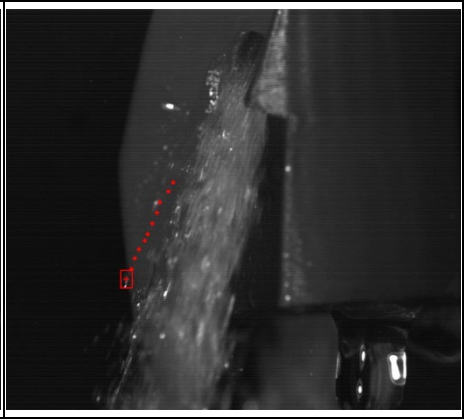
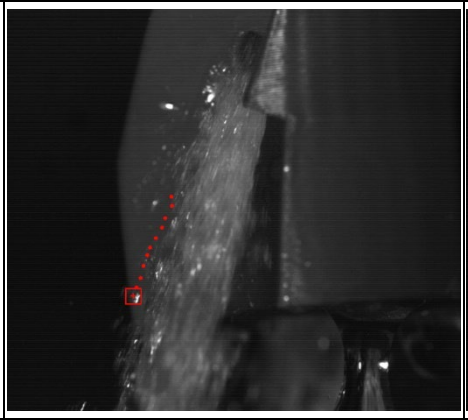
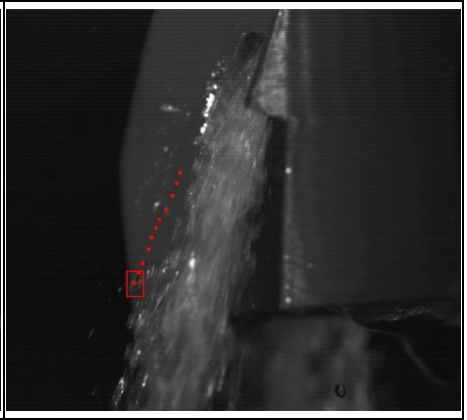
Type	Start	End	Start	End
Trajectory diagram				
Coordinates	(544.5,340.5)	(468,543)	(540,357)	(469,562)
Trajectory diagram				
Coordinates	(533.5,339)	(474,568)	(550.5,338)	(481,547)

Table 9 Pre-optimization outlet velocity measurement data (inlet pressure 0.7MPa)

Type	Start	End	Start	End
Trajectory diagram				
Coordinates	(549,301)	(450,518)	(555.5,300.5)	(460,537)
Trajectory diagram				
Coordinates	(560,264)	(478,472)	(523.5,367)	(426,572)
Trajectory diagram				
Coordinates	(520,399)	(440,608)	(536.5,346)	(442,579)

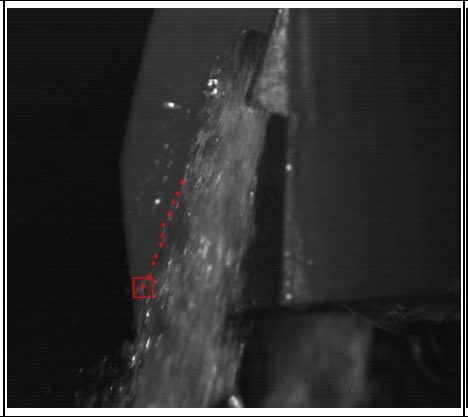
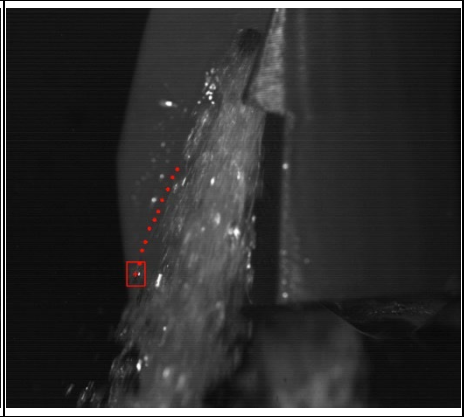
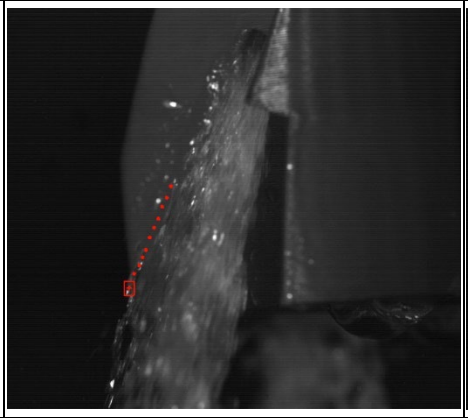
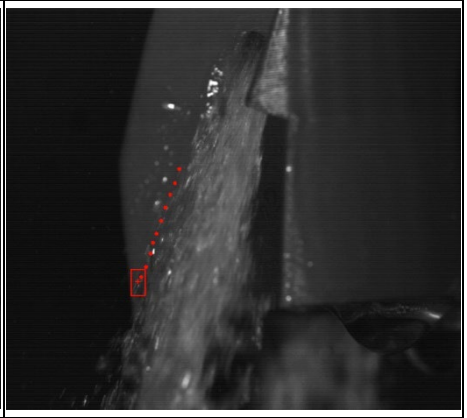
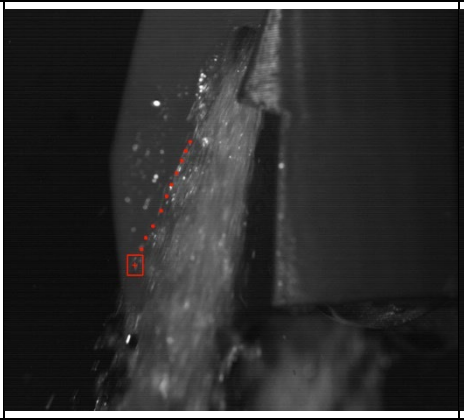
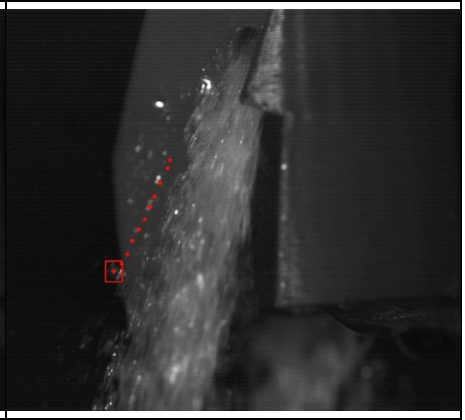
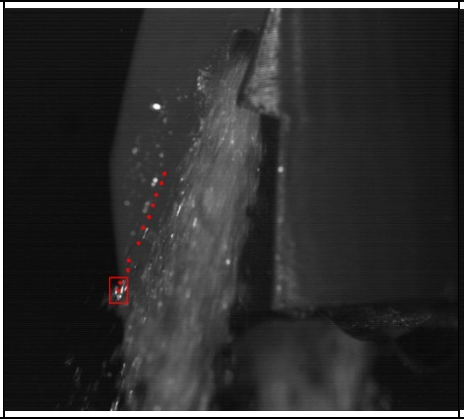
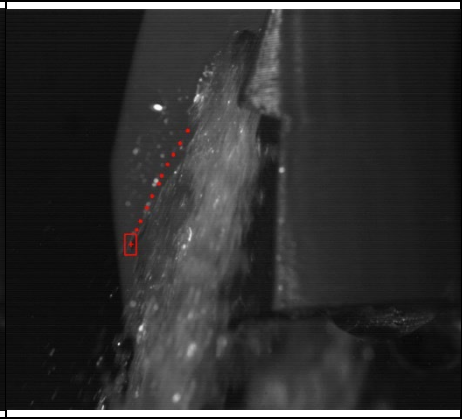
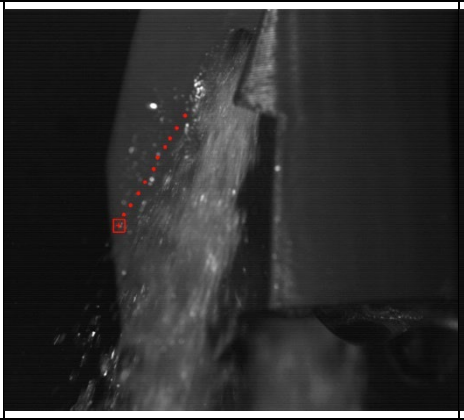
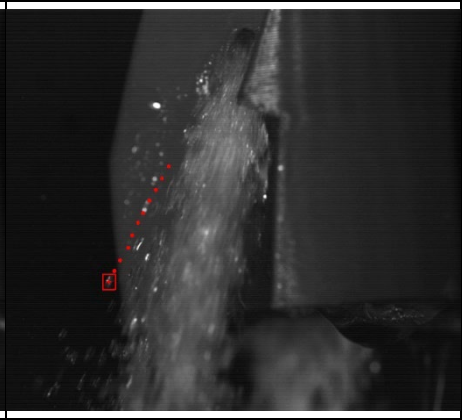
Type	Start	End	Start	End
Trajectory diagram				
Coordinates	(540.5,369.5)	(458,592)	(533.5,349)	(447,570)
Trajectory diagram				
Coordinates	(514.5,378.5)	(426,593)	(531.5,340)	(446,578)

Table 10 Pre-optimization outlet velocity measurement data (inlet pressure 0.8MPa)

Type	Start	End	Start	End
Trajectory diagram				
Coordinates	(572.5,281)	(458,537)	(525,315)	(408,546)
Trajectory diagram				
Coordinates	(516,344)	(498.6,353.2)	(560,254)	(442,490)
Trajectory diagram				
Coordinates	(567,222.5)	(430,450)	(525.5,327)	(402,567)

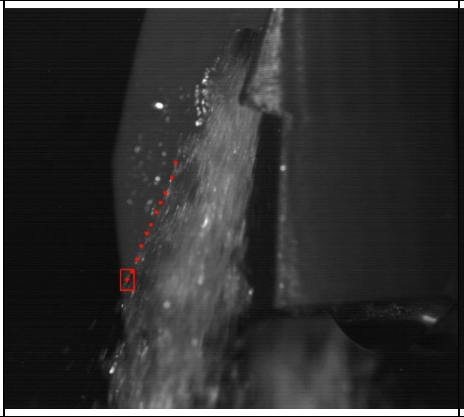
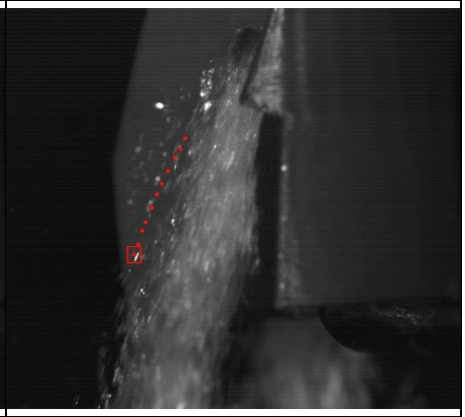
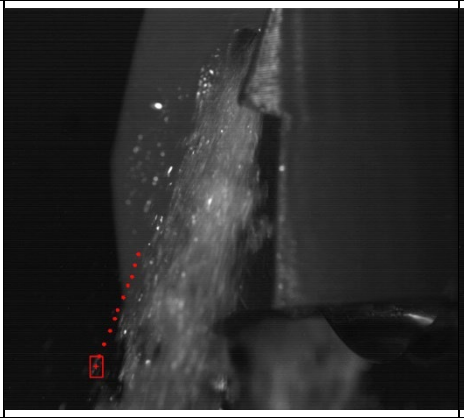
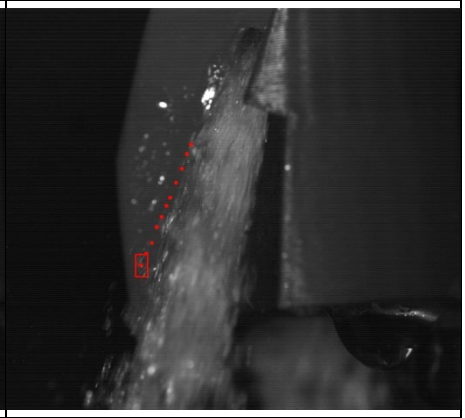
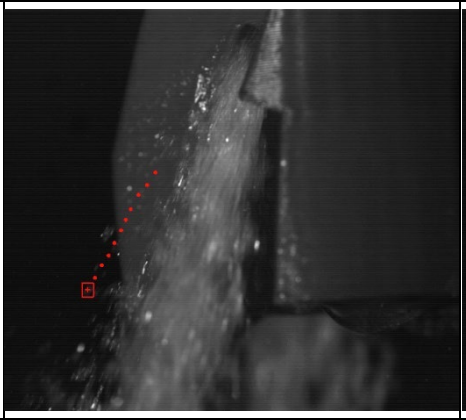
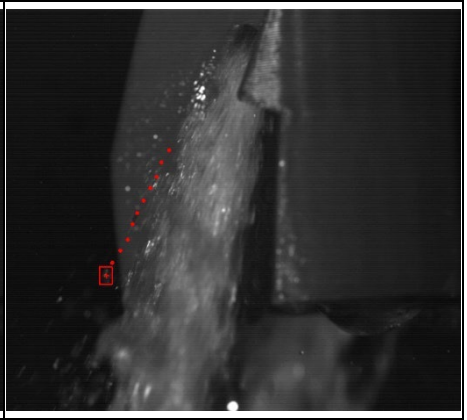
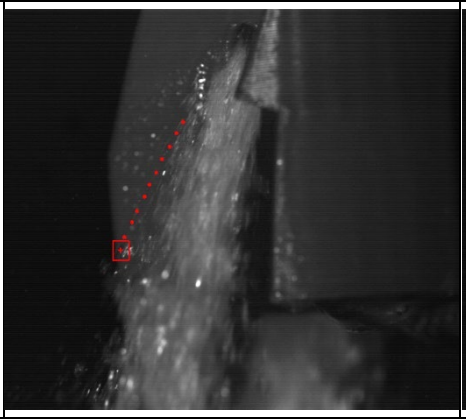
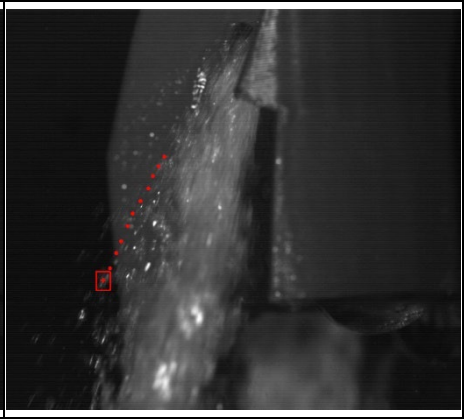
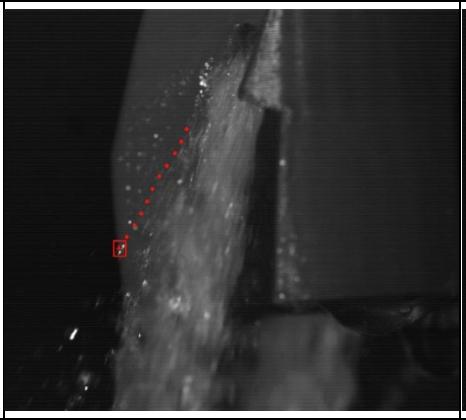
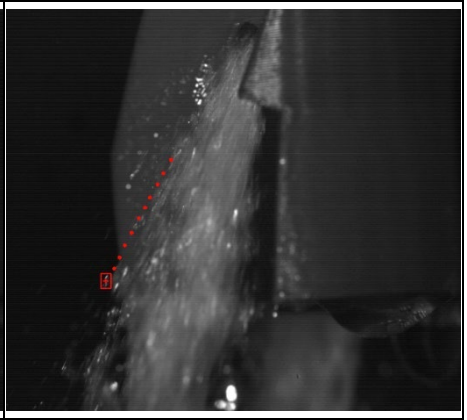
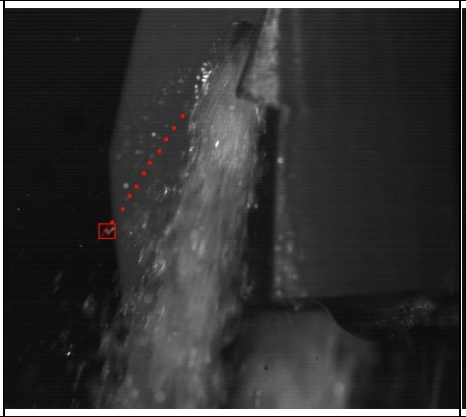
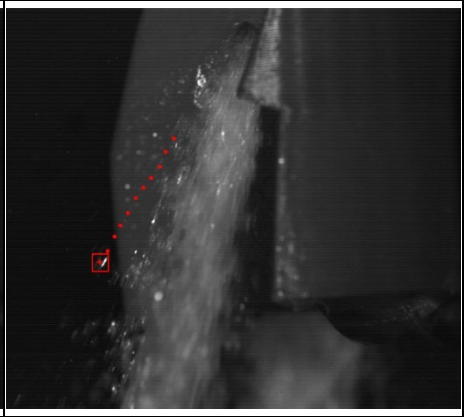
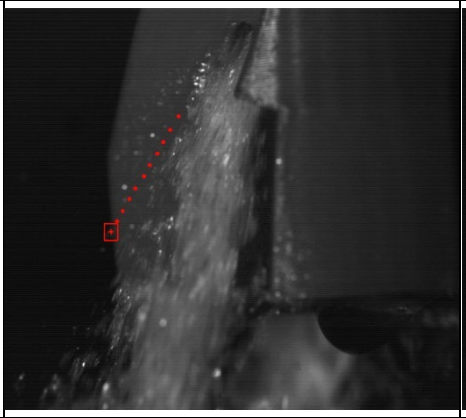
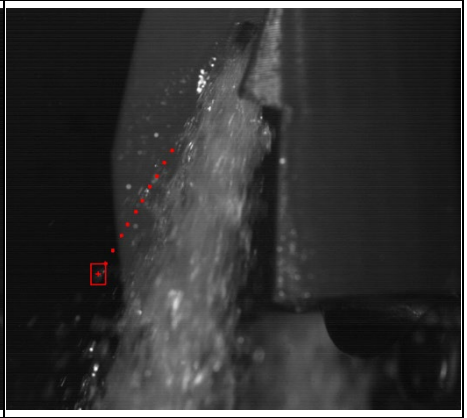
Type	Start	End	Start	End
Trajectory diagram				
Coordinates	(534.5,323)	(434,567)	(554.5,272)	(448,514)
Trajectory diagram				
Coordinates	(459,511.5)	(372,746)	(557,283.5)	(454,535)

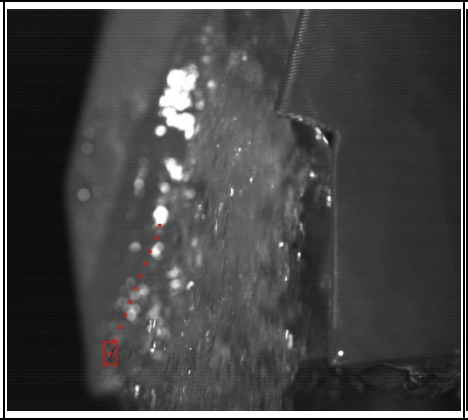
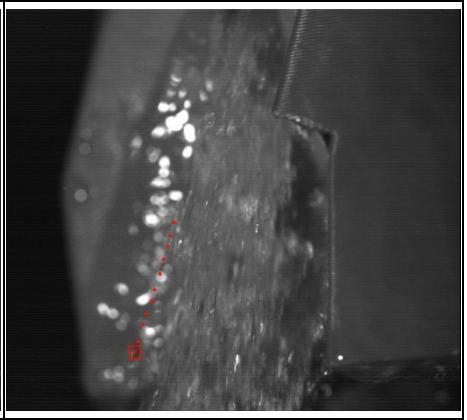
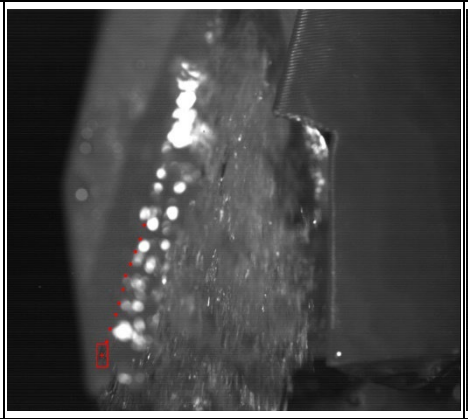
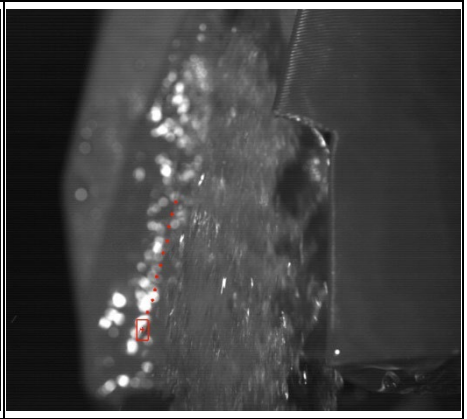
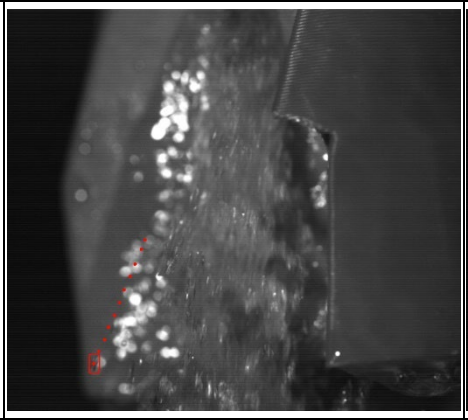
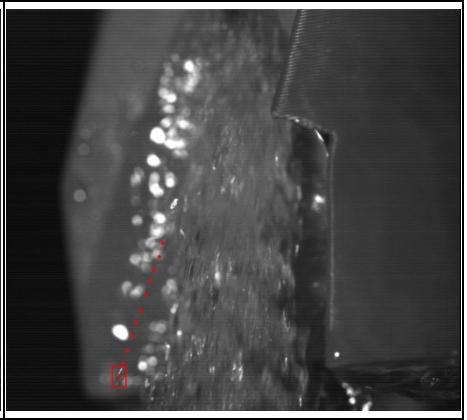
Table 11 Pre-optimization outlet velocity measurement data (inlet pressure 0.9MPa)

Type	Start	End	Start	End
Trajectory diagram				
Coordinates	(481,340)	(342,584)	(513.5,293.5)	(384,554)
Trajectory diagram				
Coordinates	(550,236.5)	(421,503)	(514.5,308)	(387,566)
Trajectory diagram				
Coordinates	(550.5,250)	(411,498)	(517.5,318.5)	(383,569)

Type	Start	End	Start	End
Trajectory diagram				
Coordinates	(546,226.5)	(389,464)	(525,271.5)	(372,530)
Trajectory diagram				
Coordinates	(542,226.5)	(402,466)	(519.5,297.5)	(368,554)

Attachment B: Optimized outlet flow velocity measurement data at different inlet pressures (1 pixel = 2.86×10^{-5} m)

Table 12 Optimized outlet velocity measurement data (inlet pressure 0.6MPa)

Type	Start	End	Start	End
Trajectory diagram				
Coordinates	(542.5,528.5)	(422,841)	(582.5,521.5)	(486,838)
Trajectory diagram				
Coordinates	(506,528.5)	(402,846)	(583,475.5)	(502,788)
Trajectory diagram				
Coordinates	(506.5,564.5)	(382,866)	(553.5,571.5)	(447,898)

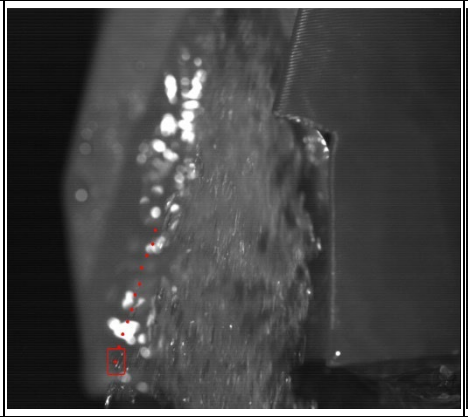
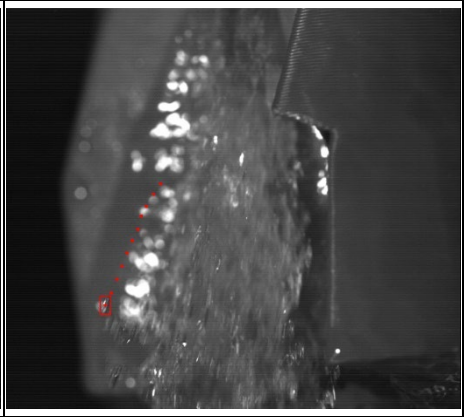
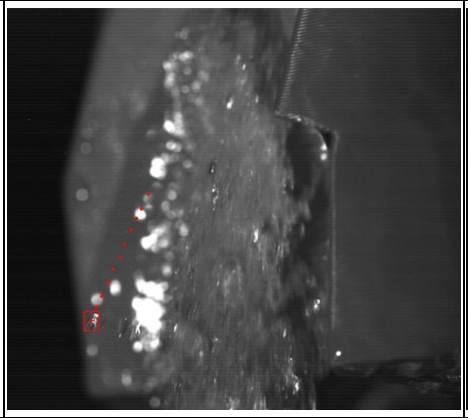
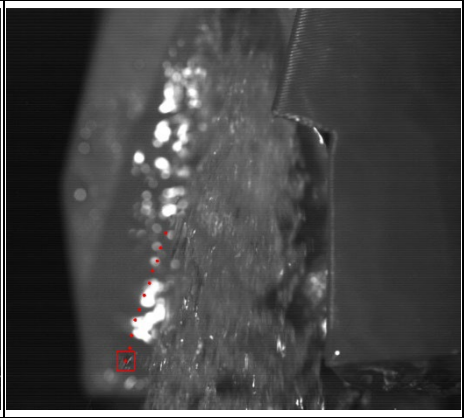
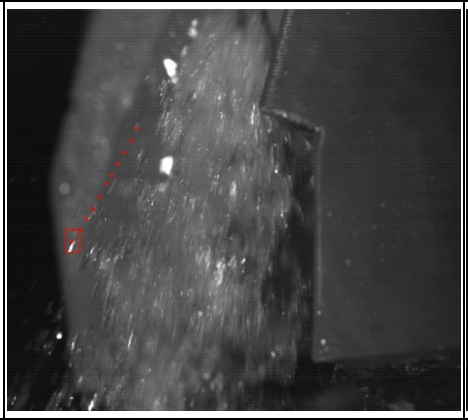
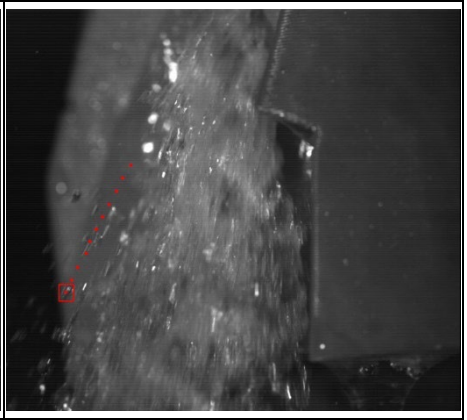
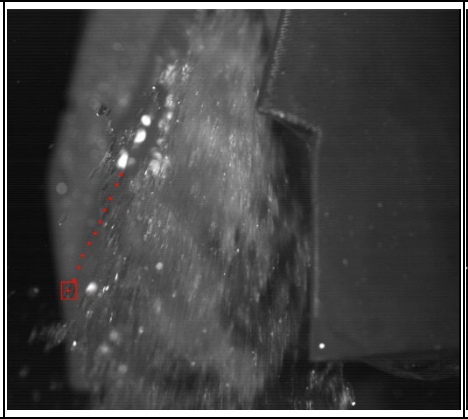
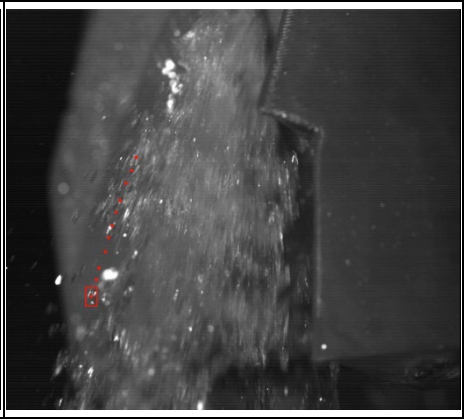
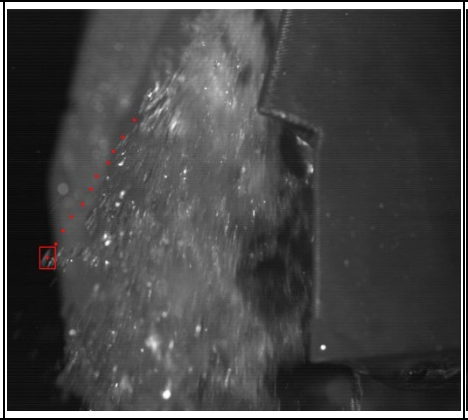
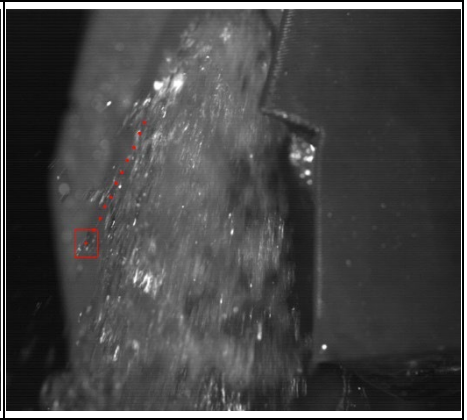
Type	Start	End	Start	End
Trajectory diagram				
Coordinates	(533,544)	(436,866)	(548.5,431)	(413,728)
Trajectory diagram				
Coordinates	(511.5,455)	(374,764)	(559,550.5)	(461,862)

Table 13 Optimized outlet velocity measurement data (inlet pressure 0.7MPa)

Type	Start	End	Start	End
Trajectory diagram				
Coordinates	(486.5,292.5)	(328,568)	(475.5,382.5)	(318,692)
Trajectory diagram				
Coordinates	(452.5,405.5)	(324,690)	(489,363.5)	(380,704)
Trajectory diagram				
Coordinates	(479.5,271.5)	(268,608)	(504.5,278.5)	(362,573)

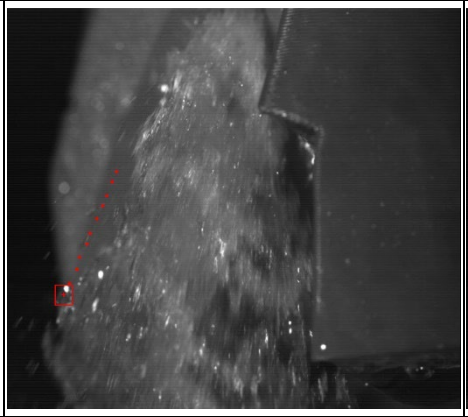
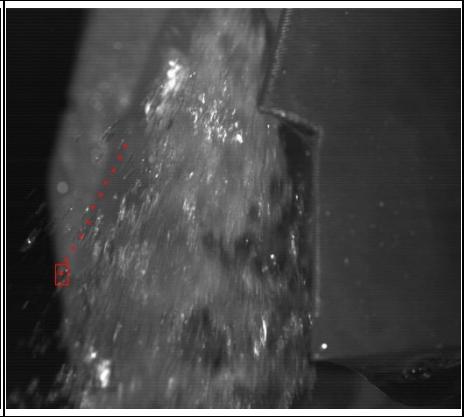
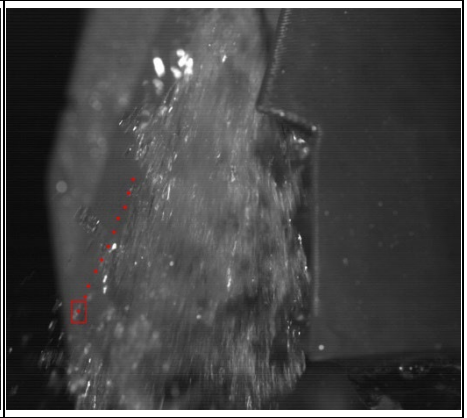
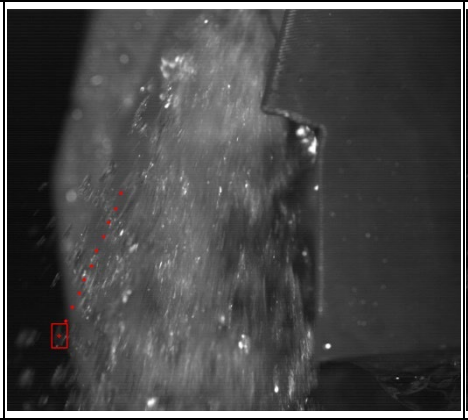
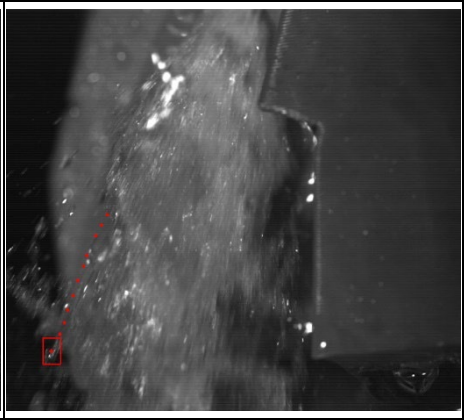
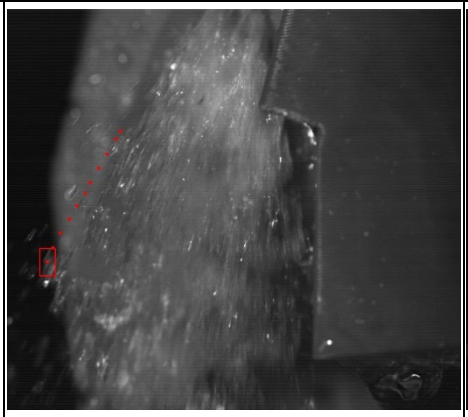
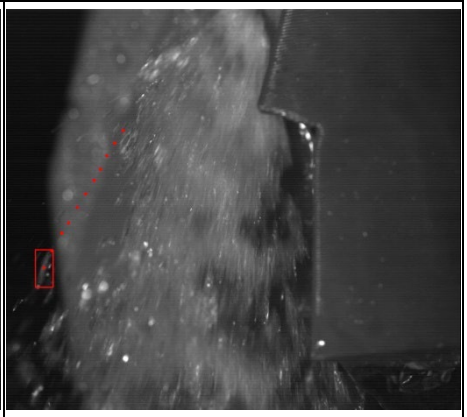
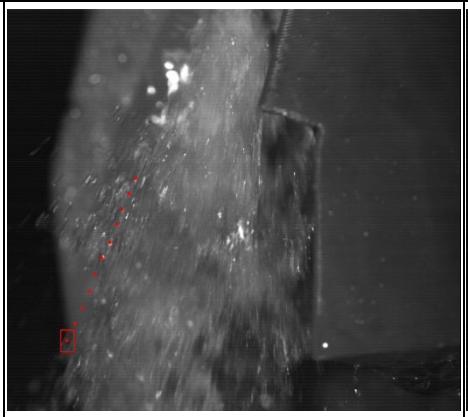
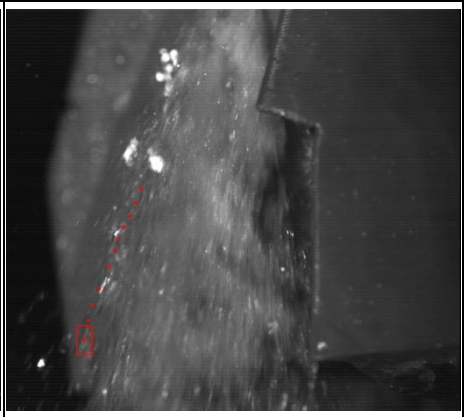
Type	Start	End	Start	End
Trajectory diagram				
Coordinates	(438.5,401)	(309,703)	(460,338)	(304,652)
Trajectory diagram				
Coordinates	(477.5,329.5)	(296,662)	(479,422.5)	(346,744)

Table 14 Optimized outlet velocity measurement data (inlet pressure 0.8MPa)

Type	Start	End	Start	End
Trajectory diagram				
Coordinates	(445.5,450)	(296,798)	(416,503)	(280,836)
Trajectory diagram				
Coordinates	(445,300.5)	(267,620)	(455.5,298.5)	(264,635)
Trajectory diagram				
Coordinates	(484.5,414.5)	(318,810)	(500.5,438.5)	(362,808)

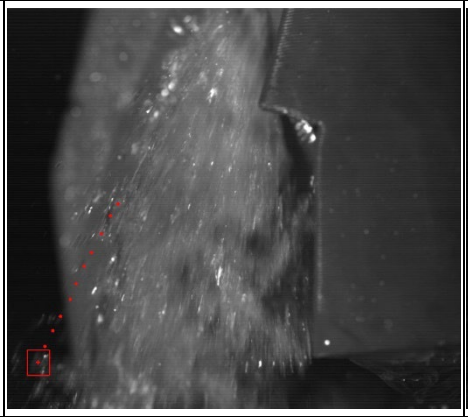
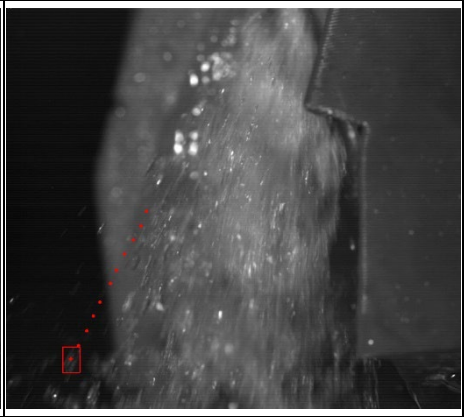
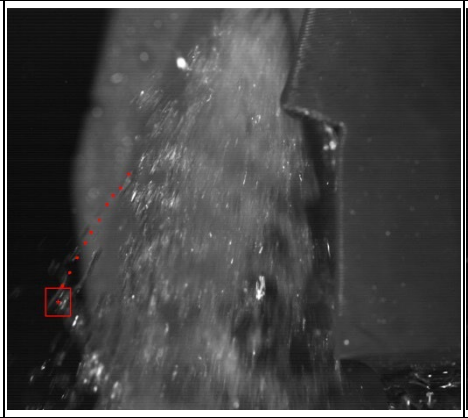
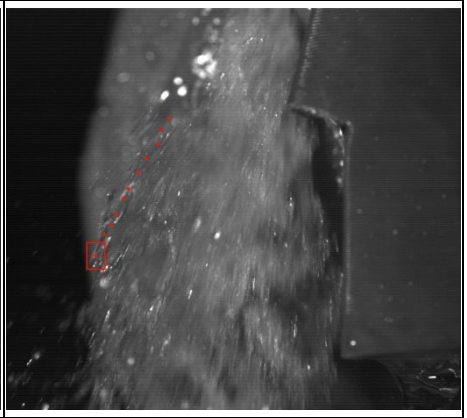
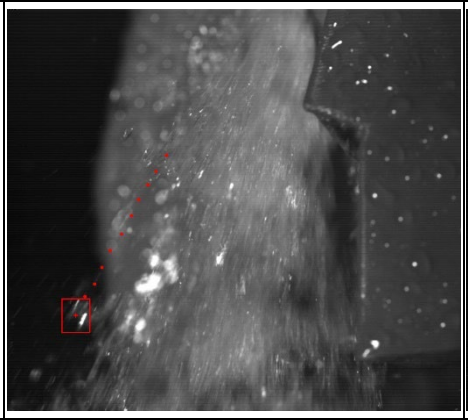
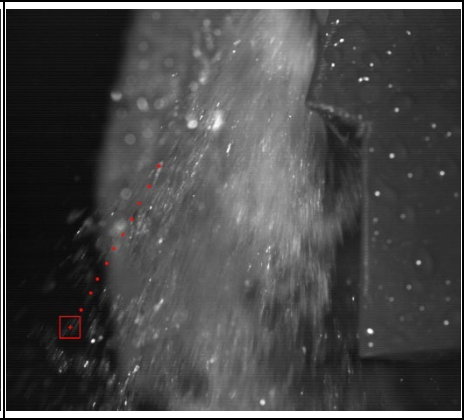
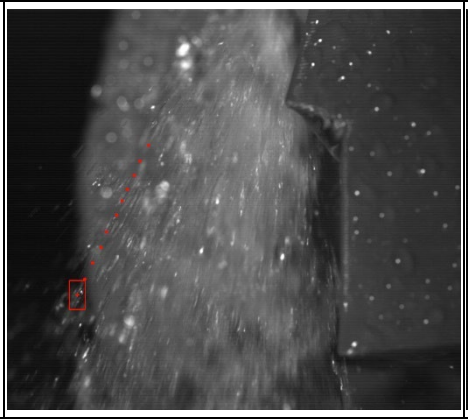
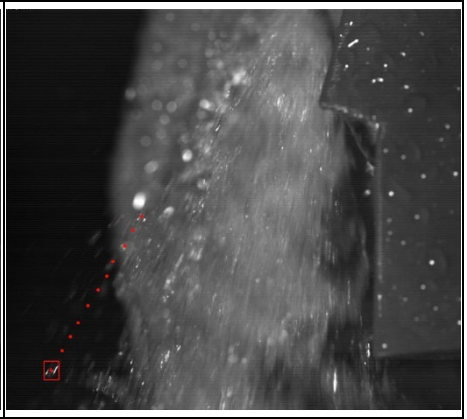
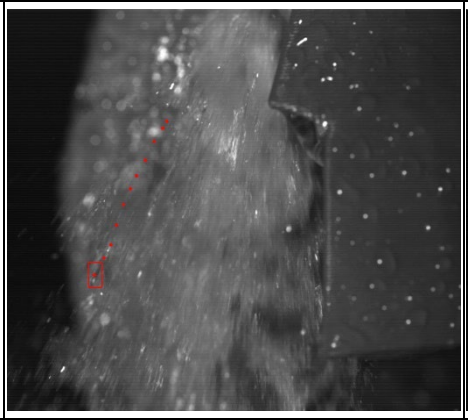
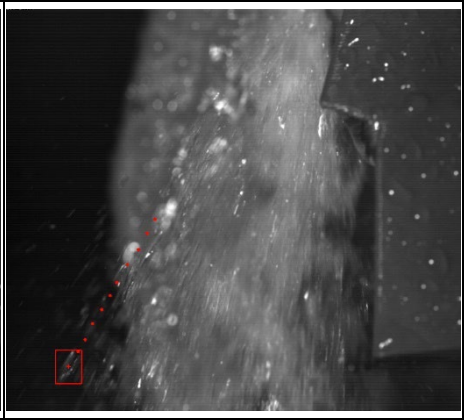
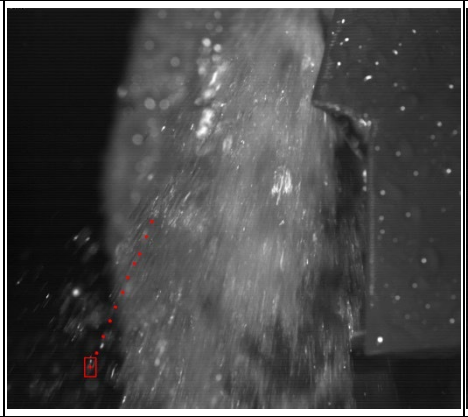
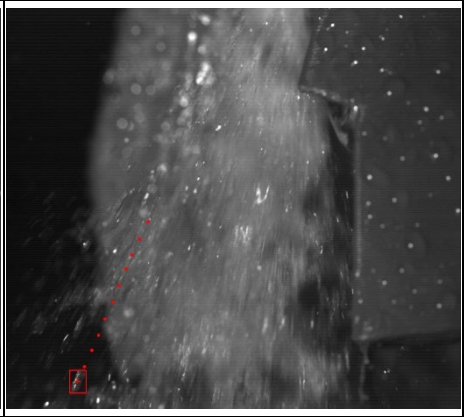
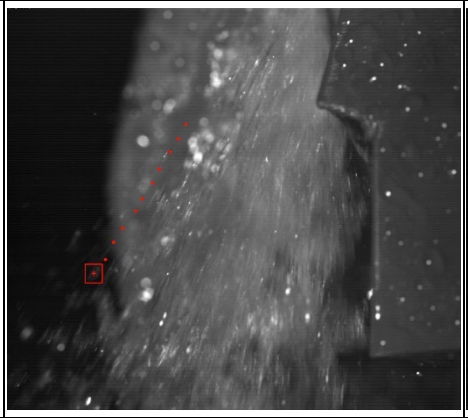
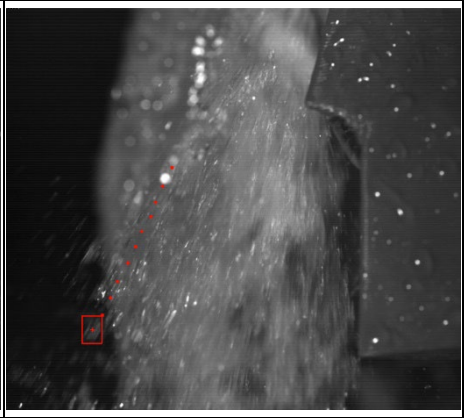
Type	Start	End	Start	End
Trajectory diagram				
Coordinates	(439,479.5)	(244,868)	(398.5,498)	(214,860)
Trajectory diagram				
Coordinates	(412,406)	(240,718)	(494,269.5)	(317,606)

Table 15 Optimized outlet velocity measurement data (inlet pressure 0.9MPa)

Type	Start	End	Start	End
Trajectory diagram				
Coordinates	(434,357.5)	(212,749)	(410,387)	(194,780)
Trajectory diagram				
Coordinates	(444.5,333.5)	(270,700)	(339.5,508)	(120,884)
Trajectory diagram				
Coordinates	(522.5,275)	(346,650)	(364.5,514)	(154,874)

Type	Start	End	Start	End
Trajectory diagram				
Coordinates	(379.5,523)	(230,879)	(407,560.5)	(236,950)
Trajectory diagram				
Coordinates	(451,285)	(226,648)	(442.5,389.5)	(248,786)